

Jesús M. Rueda-Becerril, Ph.D.

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Professional Summary

Computational physicist with a PhD in Physics and 10+ years of experience building and running simulation codes for relativistic plasmas. Author of Paramo, an open-source Fortran library that solves the Fokker-Planck equation coupled to radiative transfer. Led a 7-person upgrade of the PatchworkMHD code and benchmarked it on Frontera. Used gradient-based optimization to calibrate model parameters against observations.

Skills

Simulation: Kinetic plasma modeling (Fokker-Planck), radiative processes (synchrotron, inverse Compton), radiative transfer, relativistic MHD, hydrodynamics (SPH), Lagrangian particle dynamics, numerical PDEs, finite-difference/finite-volume methods, numerical linear algebra, parameter estimation, uncertainty quantification

HPC & Parallel: OpenMP, MPI, OpenACC (prototyped GPU offloading), SLURM, PBS, HDF5

ML & Data: Scientific ML, PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, Pandas, Astropy, random forest, SVM, neural networks, gradient-based optimization, parameter estimation

Languages: Python, C/C++, Fortran, Rust, R, SQL, Java, Julia, Shell

Tools: Git, Docker, Jenkins, QGIS, ParaView, VisIt

Experience

Independent Contractor

Jan 2026 – Present

Mercor

Remote

Physics subject-matter expert evaluating frontier language models for a leading AI lab.

- Reviewed 50+ expert-level problems in physics, hydrodynamics, astrophysics, and cosmology; wrote original prompts that stumped frontier models.

Spatial Data Scientist

May 2025 – Jan 2026

TealWaters

Seattle, WA

Built geospatial ML pipelines and optimized terrain-analysis code for wetland mapping.

- Optimized the terrain-modeling code that computes elevation derivatives from 1 m, 3 m, and 10 m DEMs over 20 tiles, improving its computational performance.
- Mapped wetland probability across the 2,160 km² Skykomish watershed with the Wetland Intrinsic Potential (WIP) random-forest model, reaching >75% accuracy.

Independent Researcher & Open-Source Developer

Jan 2024 – Present

Self-directed

Seattle, WA

Simulation research and open-source development, ongoing alongside industry roles.

- Co-developed Tleco, a Rust/Python toolkit that models radiation from relativistic outflows by solving the Fokker-Planck equation coupled to radiative transfer; published in ApJ (2024).
- Developing WindsOfChange with the UW Forest Mycobiome Lab: a C++/R library of Lagrangian particle models for turbulent transport over forested, hilly terrain, parallelized with RcppParallel and OpenMP.
- Used AI coding assistants in the most recent stages of development; changes were accepted only if they passed the unit-test suite and, for some functions, matched reference solutions.

Software Engineer

Apr 2022 – Jan 2024

Paychex

Remote

Built and tested data-integration services between enterprise systems and Oracle E-Business Suite.

- Developed and tested production data-pipeline software (Java, Python, SQL) with CI/CD.

Postdoctoral Research Associate

Feb 2021 – Apr 2022

Rochester Institute of Technology

Rochester, NY

HPC simulations of supermassive black hole binaries in an NSF-funded project.

- Led a 7-person team upgrading PatchworkMHD, a C code for general relativistic magnetohydrodynamic simulations of supermassive black hole binaries.
- Added black hole spin to the simulations with under 5% runtime overhead.
- Benchmarked new features on TACC's Frontera supercomputer (~10,000 node-hours) to find performance bottlenecks.
- Calibrated Paramo against Fermi-LAT spectra with gradient-based optimization; supervised a graduate student on the same problem.
- Co-authored a Fokker-Planck model of relativistic turbulent heating balanced by radiative cooling, with transport coefficients from PIC simulations (MNRAS 2022).

Postdoctoral Research Fellow

Purdue University

Oct 2018 – Nov 2020

West Lafayette, IN

Radiative-transfer modeling of relativistic jets (blazars and gamma-ray burst afterglows).

- Architected Paramo, an open-source Fortran 95 Fokker-Planck and radiative-transfer library; parallelized and optimized with OpenMP.
- Paramo has been adopted by other research groups, including in Combi & Siegel (2023).
- Co-I on two NASA Fermi Guest Investigator proposals; one was funded (\$68K) to model blazar radiation.
- Mentored 3 graduate students, co-authoring papers with each.

Postdoctoral Research Fellow

Universidad Michoacana de San Nicolás de Hidalgo

Jan 2018 – Sep 2018

Morelia, Mexico

- Built a Python pipeline that turned GRTrans black hole simulation images into training data for an SVM classifier of black hole spin.
- Released an open-source Python tool for computing radiative-transfer spectra and light curves in relativistic astrophysics.

Graduate Research Assistant

Universitat de València

Oct 2011 – Jul 2017

Valencia, Spain

- Studied how magnetization of shocked plasma shapes the radiation spectra of relativistic internal shocks, using the C-SPEV simulation code.
- Automated runs of the C-SPEV radiative-transfer code with Shell and Python pipelines: 10–60 simulations per project and tens of GB of HDF5 output.
- Extended C-SPEV to compute discrete and continuous spectra from particle distributions of arbitrary shape.
- Published 2 first-author papers in MNRAS.

Education

Universitat de València

Ph.D. in Physics, *Excellent Cum Laude*

Valencia, Spain

2017

Thesis: Numerical treatment of radiation processes in the internal shocks of magnetized relativistic outflows

Universidad Michoacana de San Nicolás de Hidalgo

M.Sc. in Physics

Morelia, Mexico

2011

Thesis: Study of TOV stars with the SPH method (wrote an SPH hydrodynamics code to study the stability of relativistic stars).

Projects

Blazar Population ML

Scientific ML (Python, scikit-learn, PyTorch)

Jan 2026 – Present

github.com/altjerue/ilwikak-ml

- Applying classical ML (PCA, UMAP, Gaussian mixture models, HDBSCAN) to the Fermi-LAT 4LAC-DR3 catalog (3,407 sources) to test whether jet baryon-loading constraints from my 2021 simulation study can be recovered from observables; Gaussian process regression on a 334-source MOJAVE subsample in progress.
- Planned next stage: a neural emulator of Tleco trained on synthetic spectra, then flare forecasting on Tleco-generated light curves.

Selected Publications (9 peer-reviewed articles, 3 first-author)

- Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Tleco: A Toolkit for Modeling Radiative Signatures from Relativistic Outflows*, ApJ 976, 182 (2024), arXiv:2405.17581.
- Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Balancing Turbulent Heating with Radiative Cooling in Blazars*, MNRAS 513, 5766–5779 (2022), arXiv:2201.07790.
- **Rueda-Becerril, J. M.**, Harrison, A. O., & Giannios, D. *Blazar jets launched with similar energy per baryon, independently of their power*, MNRAS 501, 4092–4102 (2021), arXiv:2009.02273.
- **Rueda-Becerril, J. M.**, Mimica, P., & Aloy, M. A. *On the influence of a hybrid thermal–non-thermal distribution in the internal shocks model for blazars*, MNRAS 468, 1169–1182 (2017), arXiv:1612.06383.